about the number system in the following Chapter.

8.3 NUMBER SYSTEM

Number system is commonly used to count any activity or articles. In practical life, we are using decimal number system. In decimal number system, 10 digits(0,1,2,3,4,5,6,7,8,9) are used. But in digital electronics, we use '1' and '0'.

Computers, microprocessor and digital electronic devices do not process decimal numbers. Instead, they work with binary number, which use only the two digits'0' and'1'.

People do not like working with binary numbers, owing to their very lengthy combinations of digits, while representing larger decimal values.

As a result, octal and hexadecimal numbers are widely used to compress long strings of binary numbers. Some number systems are given below.

8.3.1 Binary Number

Binary number contains only two numbers of '0' and '1'. It has radix or base of '2'.

Example: 1010_2

Almost all digital systems are based on binary number. A switch is one example of a natural binary device, because it exists only two states, namely ON or OFF, 1 or 0.

8.3.2 Octal Number

Octal number contains only eight numbers of 0,1,2,3,4,5,6 and 7. It has a radix or base of 8.

Example: 7612_8

8.3.3 Hexadecimal Number

Hexadecimal number contains only sixteen numbers of 0,1,2,3,4,5,6,7,8,9,A,B,C,D,E, and F. It has a radix or base of 16.

Example: $508D_{16}$

8.3.4 BCD (Binary Coded Decimal) Number

A nibble is a string of 4 bits. BCD numbers express each decimal digit as nibble. It is a decimal number represented in binary form with 0 and 1. The lowest number is 0000 (0) and the highest number is 1001(9)

Example: $1000\ 0111_{BCD}$

8.3.4.1 Place Value

The binary, octal, decimal, and hexadecimal numbers are weighted numbers. Hence, every number system can be converted into any other number system through a process called conversion. After conversion, the weight of the number should not be varied. The weight of each number is represented as follows.

8.3.4.2 Decimal Number System

Number	2	8	5	7	.4	5
Weight of each digit	10^3	10^2	10^1	10^0	$.10^{-1}$	10^{-2}

8.3.4.3 Binary Number System

Number	1	0	1	1	.0	1
Weight of each digit	2^3	2^2	2^1	2^0	$.2^{-1}$	2^{-2}

8.3.4.4 Octal Number System

Number	7	3	5	6.	3	2
Weight of each digit	8^3	8^2	8^1	8^0	$.8^{-1}$	8^{-2}

ELEC_EQUIP_Chapter08.indd 164 11/10/2020 11:50:47 AM